

Quadratic inequalities:

How to solve quadratic inequalities?

Guidelines for solving Quadratic Inequalities

1. Find all the zeros of the polynomial, and arrange the zeros in increasing order. The zeros are called its critical numbers.
2. Plot those numbers on the number line as open or closed points based upon the original inequality symbol.
3. Choose a test value in each interval to see if the interval satisfies the inequality or not. If the test value produces a true statement, the entire interval will be true. If the interval produces a false statement, the entire interval is false.
4. Clearly graph your solution and state the solution using interval notation or inequalities.

Example:

Solve the quadratic inequality $x^2 - 4x > -3$

Solution:

Step 1: Make one side of the inequality zero

$$\begin{aligned}x^2 - 4x &> -3 \\x^2 - 4x + 3 &> 0\end{aligned}$$

Step 2: Factor the quadratic expression

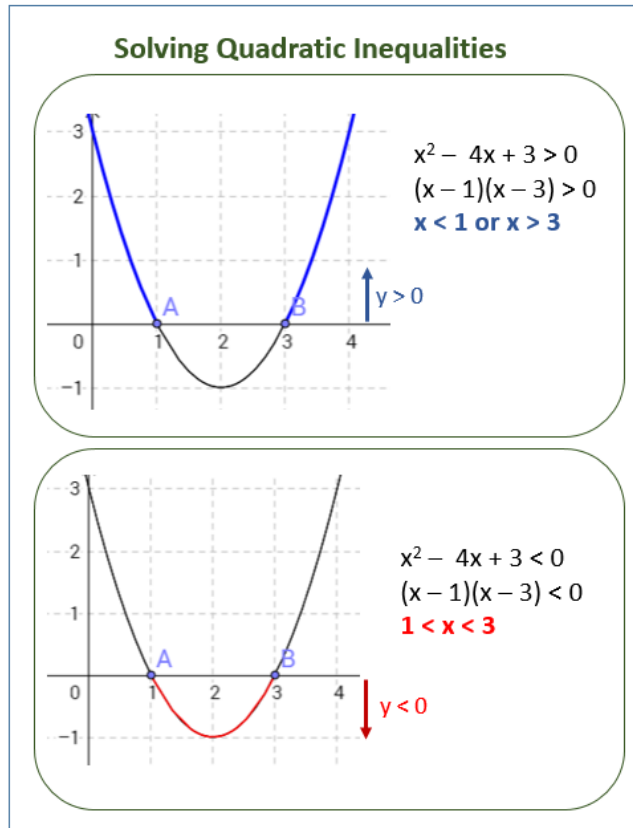
$$\begin{aligned}x^2 - 4x + 3 &> 0 \\(x - 3)(x - 1) &> 0\end{aligned}$$

Step 3: Find the range of values of x which satisfies the inequality.

$(x - 3)(x - 1) > 0$ (y is positive): we choose the interval for which the curve is above the x -axis.
 $x < 1$ or $x > 3$

Note: If the quadratic inequality was $(x - 3)(x - 1) < 0$ (y is negative), we would have chosen the interval for which the curve is below the x -axis i.e. $1 < x < 3$

The following graphs show the solutions for $x^2 - 4x + 3 > 0$ and $x^2 - 4x + 3 < 0$. Scroll down the page for more examples and solutions.



Example:

Solve $2x^2 < 9x + 5$

Solution:

Step 1: Make one side of the inequality zero

$$2x^2 < 9x + 5$$

$$2x^2 - 9x - 5 < 0$$

Step 2: Factor the quadratic expression

$$2x^2 - 9x - 5 < 0$$

$$(2x + 1)(x - 5) < 0$$

Step 3: Find the range of values of x which satisfies the inequality.

$(2x + 1)(x - 5) < 0$ (y is negative): we choose the interval for which the curve is below the x -axis.

$$-\frac{1}{2} < x < 5$$

Note: If the quadratic inequality was $(2x + 1)(x - 5) > 0$ (y is positive) we would have chosen

the interval for which the curve is above the x -axis i.e. $x < -\frac{1}{2}$ or $x > 5$

Examples:

1. Solve $x^2 - 6x - 16 \leq 0$
2. Solve $2x^2 - 11x + 12 > 0$
3. Solve $x^2 + 4 > 0$
4. Solve $x^2 - x - 12 \leq 0$
5. Solve $2x^2 + 3x - 5 > 0$